Simplify algebraic expressions, expand binomial products and factorise monic quadratic expressions

Learning intention

- simplify algebraic expressions, expand binomial products and factorise monic quadratic expressions.

Teach and model

- using manipulatives such as algebra tiles or area models to expand or factorise algebraic expressions with readily identifiable binomial factors.
- recognising the relationship between expansion and factorisation, and using digital tools to systematically explore the factorisation of x^2+mx+n where m and n are integers.
- expanding combinations of binomials such as $(x+7)(x+8),\;(x+7)(x-8),\;(x-7)(x+8),\;(x-7)(x-8)$ to identify expansion and factorisation patterns related to $(x+a)(x+b)\;=\;x^2+(a+b)x+ab$, where a and b are integers.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.